Daniela Betancurt-Anzola

Microbial ecology | Multi-omics | Experimental microbiology

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Microbial ecologist interested in how metabolic and genetic interactions shape microbial communities, their functions, and their responses to disturbance. My doctoral research investigates carbon-cycling potential in Colombia's tropical Andean wetland soils. I lead a multi-omics project from field sampling and laboratory processing through genome-resolved metagenomics, amplicon sequencing, untargeted metabolomics, and statistical analysis. I also developed an HPC workflow to reconstruct genomes from uncultivated microorganisms and investigate their metabolic capabilities. Previously, I combined microbial genetics, automated strain screening, and mouse experiments to develop and evaluate a candidate probiotic strategy for reducing dietary mercury bioavailability. I am interested in bringing computational and experimental approaches together to test how interactions among microorganisms influence community function.

Education

2022–present **Ph.D. Student, Biochemistry, Microbiology and Molecular Biology**, The Pennsylvania State University, State College, PA

- Expected completion: Summer 2027.

Advisor: Estelle Couradeau, Ph.D. (2024–present).

2021–2022 **M.Sc. Computational Biology**, Universidad de los Andes, Bogotá, Colombia

- Thesis: Divergent response of the gut microbiome to a high-fiber diet.

Advisors: Alejandro Reyes, Ph.D., and Federico Rey, Ph.D.

2012–2018 **B.Sc. Microbiology and B.Sc. Biology; Minor in Psychobiology**, Universidad de los Andes, Bogotá, Colombia

- Thesis: Effects of exposure to *P. aeruginosa* PAO1 on the behavior of zebrafish *Danio rerio*.

Advisors: Veronica Akle, Ph.D., and Martha Vives, Ph.D.

Research Experience

2024–present **Graduate Researcher, Couradeau Lab.**, The Pennsylvania State University, State College, PA

- Co-designed and led a multi-omics project from field sampling through computational analysis, interpretation, and publication, coordinating the collection and processing of soil samples, sample metadata, DNA and metabolite extractions, untargeted GC-MS/MS and LC-MS/MS analyses, and permits to transport plant material from Colombia to the United States.
- Developed Bash, Python, and R scripts and SLURM job arrays for an HPC pipeline spanning raw-read QC, assembly, mapping, binning, quantification, gene prediction, taxonomic classification, and functional annotation.
- Integrated amplicon, gas-flux, biogeochemistry, metagenomic, and metabolomic datasets using regression, multivariate analysis, mixed-effects models, clustering, and dimensionality reduction; process untargeted metabolomics data with MZmine.
- Apply genome-resolved metagenomics to investigate abundant uncultivated microbial lineages and carbon-cycling potential in tropical wetland soils.

2022–2024 **Graduate Researcher, Bisanz Lab.**, The Pennsylvania State University, State College, PA

- Designed constructs, cloning strategies, assays, strain-selection criteria, and experimental controls for a candidate probiotic developed to reduce dietary mercury bioavailability in the mammalian gut.
- Screened hundreds of bacterial strains across a mercury-concentration gradient using Opentrons; wrote and modified Python protocols for Opentrons.
- Contributed across the complete mouse-study workflow including study design, dosing, animal handling, sample and tissue collection, data analysis, and interpretation, performed toxicity experiments to establish appropriate dosage levels, and used preliminary findings to select mercury concentrations, bacterial strains, and microbial communities for subsequent studies.
- Developed qPCR protocols to assess the effectiveness of the computational tool StrainR2 developed in the lab.
- Mentored four undergraduate researchers, supervised timelines, assigned work, reviewed code and analyses, and provided feedback.

2020–2022 **Research Assistant, Computational Biology and Microbial Ecology Lab.**, Universidad de los Andes, Bogotá, Colombia

- Integrated gut amplicon, metabolomics, and biomarker data from a mouse dietary-fiber experiment using QIIME2, R, and Python.

2021–2022 **Laboratory Manager, Affect and Cognition Lab.**, Cornell University, Ithaca, NY

- Managed molecular biology workflows, developed PCR protocols, genotyped human saliva samples, and analyzed gut metagenomic data in relation to physiological and behavioral measurements.
- Wrote SOPs, protocols, documentation, and training materials while working under IACUC, IRB, biosafety, and controlled-access human-data requirements.
- Mentored two undergraduate students, coordinated project timelines and assignments, reviewed analyses, and provided feedback.

2018–2019 **Research Assistant, Affect and Cognition Lab.**, Cornell University, Ithaca, NY

- Completed bench-to-bioinformatics profiling of the human gut microbiome, including DNA extraction, purification, normalization, PCR optimization, and 16S rRNA gene sequencing of human stool samples.
- Trained Long–Evans rats, recorded brain data, and assisted with brain surgeries for studies of memory, aging, and dopamine decline.

Additional Research Experience

2018 **Summer Research Intern, Neurobiology of Memory and Learning Lab.**, Cornell University, Ithaca, NY

2017–2018 **Undergraduate Researcher, Neuroscience and Circadian Rhythms Lab.**, Universidad de los Andes, Bogotá, Colombia. Evaluated the effect of *P. aeruginosa* PA01 on the behavior of zebrafish using EthoVision

2013 **Undergraduate Researcher, Mycology and Phytopathology Lab.**, Universidad de Los Andes, Bogotá, Colombia. Developed a protocol for the preparation of permanent fungi demonstrative slides for the laboratory class. Maintained the fungi collection of the lab.

Publications

Betancurt-Anzola, D., Freire-Zapata, V., Villa, J. A., Robles Zaragoza, Cala, M., L., Tfaily, M., Santamaría Vanegas, & Couradeau, E. Deciphering Microbiome–Metabolome Relationships in the Chingaza páramo Through Multi-Omics Integration **(In preparation).**

Betancurt-Anzola, D., Glass, S., J. A., Santamaría Vanegas, & Couradeau, E. Host Identity and Soil–Plant Compartments Structure Microbial Communities in Endemic Páramo Plants **(In preparation).**

Betancurt-Anzola, D.*, Robles Zaragoza, L.*, Glass, S., Machado, J.-L., J. A., Santamaría Vanegas, & Couradeau, E. Restructuring of Páramo Soil Microbiomes Under Contrasting Temperature and Moisture Regimes **(In preparation).**

Betancurt-Anzola, D.*, Robles Zaragoza, L.*, Tami-Barrera, L., Buytaert, W., Calispa, M., Castillo-Figueroa, D., Diazgranados, M., Espinosa-Ramirez, A. J., Gallery, R. E., Gerard, F., Gutiérrez-Lagoueyte, M. E., Hofstede, R., Hurtado-M, A. B., Lasso, E., Longo Sánchez, M. C., López-Alvarez, D., López-Uribe, M. M., Morales-Rozo, C., Rocha Cárdenas, D. S., Ruiz-Carrascal, D., Sanchez, A., Scott, C. A., Suarez Ballesteros, C. I., Suarez, E., Tovar, C., Villa, J. A., Vuille, M., Rosero-Cuastumal, E. P., Benavides Bolanos, J. A., Santamaría Vanegas, J., Machado, J.-L., Maximova, S., & Couradeau, E. The páramos' invisible collapse: Integrating soil carbon, water, and governance for Andean sustainability. **(In preparation).**

Betancurt-Anzola, D., Vanegas, J., Jurgensen, S. K., Wrighton, K. C., Santamaría-Vanegas, J., & Couradeau, E. (2026). Genome-resolved metagenomics reveals a high diversity of novel, non-methanogenic lineages in an Andean páramo wetland. ISME Communications. doi:10.1093/ismeco/ycag294

Couradeau, E., Vanegas, J., **Betancurt-Anzola, D.**, Glass, S., Eckert, K., Bechtold, E. K., Ellenbogen, J. B., Robles Zaragoza, L., Wrighton, K., & Santamaría Vanegas, J. (2026). Soil microbial diversity in páramos wetland of the Colombian Andes reveals novel and unique features within a global wetland database. Microbial Ecology. doi:10.1007/s00248-026-02738-x

Heber, K., Tian, S., **Betancurt-Anzola, D.**, Koo, H., & Bisanz, J. E. (2025). StrainR2 accurately deconvolutes strain-level abundances in synthetic microbial communities. Bioinformatics, 41(8), btaf440. doi:10.1093/bioinformatics/btaf440

Pérez-Quintero, A. L., Rodriguez-R, L. M., Cuesta-Morrondo, S., Hakalová, E., **Betancurt-Anzola, D.**, Valera, L. C. C., Cardenas, L. A. C., Matiz-Céron, L., Jacobs, J. M., Roman-Reyna, V., Muñoz, A. R., Giraldo, A. J. B., & Koebnik, R. (2023). Comparative genomics identifies conserved and variable TAL effectors in African strains of the cotton pathogen Xanthomonas citri pv. malvacearum. Phytopathology, 113(8), 1387–1393. doi:10.1094/PHYTO-12-22-0477-SC

*Equal contribution.

Research Presentations

2026 **Poster, One Health Microbiome Center Symposium**, The Pennsylvania State University, University Park, PA.
Abundant uncultivated lineages lacking methanogenic potential dominate carbon-cycling genes in Andean páramo soils

2026 **Talk, Penn State Water Conference,** The Pennsylvania State University, University Park, PA.
Abundant uncultivated lineages lacking methanogenic potential dominate carbon cycling genes in Andean páramo soils

2026 **Poster, JGI User Meeting Integrating Experimentation with Data**, Seattle, WA.
Genomic Dark Matter Drives Metabolic Innovation in High-Altitude Tropical Andean Wetlands

2025 **Poster, Páramos 2025 Symposium**, Bogotá, Colombia.
Microbial Community Structure and Function Along Topographically Driven Microenvironments in a Tropical Andean Peatland; **Best Poster Award**

2025 **Poster, Decoding the Biosphere Symposium**, New England Biolabs, Ipswich, MA.
Microbial Community Structure and Function Along Topographically Driven Microenvironments in a Tropical Andean Peatland

2023 **Rapid-fire talk and poster, ASM Microbe**.
Understanding and Manipulating Dietary Metal Bioavailability through the Gut Microbiome

2020 **Invited Speaker, North-South-South Network Xanthomonads Genomics Course**.
Sequencing Methods An Overview

2018 **Poster, V Biological Sciences Congress**, Universidad de los Andes.
Undergraduate thesis: Effects of the exposure to *P. aeruginosa* PA01 on the behavior of zebrafish *Danio rerio*

Workshop Instruction

2024–2026 **Workshop Chair, Data Analysis Working Group**, One Health Microbiome Center, The Pennsylvania State University. Offered four workshops per semester to the microbiome center community for two years.

2024 **Workshop Lead, National Microbiome Data Collaborative**, One Health Microbiome Center, State College, PA

2024 **Workshop Lead, National Microbiome Data Collaborative**, Joint Genome Institute User Meeting, Walnut Creek, CA

2023 **Teaching Assistant, KickStart Microbiome Workshop**, One Health Microbiome Center, State College, PA

Awards and Fellowships

2025 **Interdisciplinary Innovation Fellowship**, One Health Microbiome Center, The Pennsylvania State University. Awarded $5,000 USD

2025 **Best Poster Award**, Páramos 2025 Symposium, Bogotá, Colombia

2024 **International Graduate Education Award**, The Pennsylvania State University. Supported up to six months of thesis research in the Colombian páramos. Awarded $14,000 USD

2023–2025 **Future Leaders Mentoring Fellowship**, American Society for Microbiology

2018 **Summer Undergraduate Research Fellowship (SURF)**, Universidad de Los Andes.
Research placement at Cornell University

Teaching Experience

2023 **Graduate Teaching Assistant, Molecular Genetics Laboratory**, The Pennsylvania State University. Spring and Fall 2023.

2015 **Undergraduate Teaching Assistant, Cell Biology Laboratory**, Universidad de los Andes. Fall 2015.

Leadership and Service

2025–2026 **Organizing Committee Member, One Health Microbiome Center Symposium**, The Pennsylvania State University

2024–2026 **Workshop Chair, Data Analysis Working Group**, One Health Microbiome Center, The Pennsylvania State University

2025 **Champion**, National Microbiome Data Collaborative (NMDC)

2024 **Ambassador**, National Microbiome Data Collaborative (NMDC)

2023 **First-Year Representative, Graduate Student Association**, BMMB Program, The Pennsylvania State University

2022 **Microbiome Outreach Instructor**, State College Area High School ESOL Program

Press Releases

2023 **Medical news today.** Could an experimental probiotic help detoxify mercury from diet? https://www.medicalnewstoday.com/articles/could-an-experimental-probiotic-help-detoxify-mercury-from-diet

2023 **American Society for Microbiology** A probiotic could help mitigate mercury absorption in the gut. https://asm.org/Press-Releases/2023/June/How-Microbes-May-Help-Mitigate-Mercury-Absorption

Skills and Methods

- **Programming and infrastructure:** R; Python; Bash; Git/GitHub; HPC; SLURM job arrays; Conda; FASTQ, SAM, and BAM processing
- **Metagenomics and amplicon analysis:** FastQC; MultiQC; fastp; Trimmomatic; Nonpareil; metaSPAdes; metaQUAST; Bowtie2; CoverM; MetaBAT2; MaxBin2; CONCOCT; DAS Tool; CheckM2; GUNC; dRep; fastANI; GTDB-Tk; DRAM2; HMMER; QIIME2; DADA2; MEGAN; DIAMOND
- **Multi-omics and statistics:** MZmine; untargeted GC-MS/MS and LC-MS/MS; NCBI; GTDB; KEGG; dbCAN; regression; multivariate analysis; mixed-effects models; clustering; dimensionality reduction; visualization
- **Molecular and microbial methods:** DNA extraction, purification, and normalization; PCR; qPCR; cloning; microbial culture; strain engineering and screening; BONCAT-FACs; flow cytometry; human, animal, and environmental sample processing
- **Automation, field, and animal research:** Opentrons Python protocols; liquid-handling automation; high-throughput screening; soil and wetland sampling; mouse models; rat and zebrafish studies
- **Documentation and compliance:** SOPs; IACUC; IRB; biosafety; controlled-access human data
- **Languages:** Spanish (native); English (advanced); French (intermediate); German (basic)